

# Nikhil Thankasala

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## Experience

### Charge Robotics

Robotics Software Engineering Intern · San Leandro, CA

*June 2026 to September 2026*

- Redesigned and owned the safety system for a multi-arm autonomous solar installation robot as one of five software engineers, spanning the stack from the safety controller to the operator interface.
- Shipped a production EtherNet/IP safety driver decoding gate and emergency-stop interlock chains into live safety states, making the autonomy software e-stop and pause aware.
- Streamed safety, fault, and task-state telemetry into InfluxDB with Grafana dashboards for cell uptime and downtime.
- Developed software and firmware for a servo-driven screwdriver end-effector over CANopen, replacing a commercial fastening tool in production with a two-phase speed-to-torque fastening sequence.
- Built a real-time fasten-phase detection pipeline for error-aware fault handling and recovery on every fastening cycle.
- Built process quality indicator (PQI) monitoring for all IO-Link sensors, reporting health through the robot heartbeat.

### Ambi Robotics

Robotics Intern · Berkeley, CA

*Fall 2025 to June 2026*

- Debugged, tested, and iterated on CARGO, the machine learning stacking policy behind AmbiStack, an AI-powered gantry robot that densely palletizes and containerizes mixed e-commerce parcels for logistics operations.
- Analyzed point clouds, segmentation masks, and neural network inference logs in Rerun and Grafana to investigate item crush and stack stability failures in the sim-to-real reinforcement learning system.
- Traced perception, grasp-planning, and motion-control faults to root cause across AmbiSort parcel sortation and AmbiStack state-machine pipelines, validating fixes in Docker with research scientists and engineers.

## Projects

### FORTIS: Tokamak Inspection Robot

Software, Simulation & Vision Lead

*Senior Capstone, sponsored by General Atomics (DIII-D) · [github.com/nthankas/FORTIS](https://github.com/nthankas/FORTIS)*

- Played an integral role in the design of the entire tethered inspection robot for the DIII-D tokamak and owned its full software stack, leading architecture, simulation, and perception on a five-person cross-functional team.
- Ran Monte Carlo simulations in NVIDIA Isaac Sim to evaluate drivetrain and arm stability, driving the robot's switch to an omnidirectional X-drive.
- Built the ROS 2 manipulation stack with motion planning and inverse kinematics for a custom 4-DOF arm carrying camera and thermocouple payloads.
- Developed state estimation with an Extended Kalman Filter fusing wheel odometry, IMU, and visual-inertial odometry on a Jetson Orin Nano.
- Built a five-camera perception pipeline in Python with multi-camera depth fusion, voxel mapping with cross-run change detection, and YOLOv8 object detection using ONNX inference.

### Autonomous Sorting Robot

Software Lead · Embedded C, PIC32

*Mechatronics*

- Architected a hierarchical state machine with per-zone sub-machines coordinating line-following, color classification, and servo-driven block manipulation for a fully autonomous mobile robot.
- Built a signal-conditioning pipeline for ultrasonic ranging using band-pass filtering, outlier rejection, and distance-compensated thresholding to suppress multipath reflections and angular distortion.
- Calibrated optical color sensing for reliable RGB discrimination under variable lighting, tuning thresholds empirically and constraining sensing geometry for repeatable readings.

## Technical Skills

**Languages:** Python, C, C++, MATLAB, Bash, Verilog

**Machine Learning & Perception:** PyTorch, TensorFlow, ONNX, YOLOv8, OpenCV, NumPy, stereo depth, Matplotlib

**Robotics & Simulation:** ROS 2, NVIDIA Isaac Sim, inverse kinematics, motion planning, EKF state estimation, visual-inertial odometry, Monte Carlo simulation, sensor fusion

**Embedded & Industrial:** EtherNet/IP, CANopen, IO-Link, UART, SPI, I2C, STM32, ESP32, PIC32, motor control

**Tools:** Git, Linux, Docker, Rerun, Foxglove, Grafana, InfluxDB

## Education

### University of California, Santa Cruz

Santa Cruz, CA

M.S. Electrical & Computer Engineering (Robotics & Controls), in progress

Expected June 2027

B.S. Robotics Engineering, Minor in Electrical Engineering

June 2026